Lecture 17

Linear systems and elimination

Uses.

Summary 1: Equivalent statements have the same truth value; a system requires all its equations simultaneously.
Summary 2: Real arithmetic permits division by a nonzero scalar.
Summary 12: Ordered lists record components, with equality component by component.
Summary 15: Complex arithmetic has the same associative and distributive laws, with a reciprocal for every nonzero number.

Objectives. *Elimination replaces equations by equivalent ones until the unknowns can be read in terms of free parameters. We prove why the row operations preserve every solution and why the procedure stops. A single parameter family then gives a unique solution, a family of solutions or a contradiction. The independence of the pivot count from the reduction route is a separate cited result.*

By the end of this lecture you will be able to:

- Encode linear systems as augmented matrices over the real or complex numbers.
- Prove solution preservation and carry out elimination using only nonzero pivots for division.
- Apply the proved pivot criterion to classify and parameterise all solutions.
- State which route-independence claim uses the cited uniqueness of reduced form.

17.1 Systems and their coefficient arrays

Choose $K = \mathbb{R}$ or $K = \mathbb{C}$ and keep that choice fixed. All coefficients, unknowns and parameters below belong to this number system. For integers $m, n \geq 1$, a *linear system* of m equations in n unknowns has the form

$$\sum_{j=1}^{n} a_{ij}x_j = b_i, \qquad i = 1, \dots, m. \tag{17.1}$$

The coefficients $a_{ij} \in K$ and right sides $b_i \in K$ are given. The unknowns are $x_1, \dots, x_n \in K$. The first index labels an equation and the second labels an unknown. A *solution* is an ordered list satisfying every equation. A system is *consistent* if it has at least one solution, and *inconsistent* if it has none.

A *matrix* is a rectangular array of scalars. The coefficient matrix A has m rows and n columns, with entry a_{ij} in row i and column j. The *augmented matrix* records that array together with the right sides:

$$[A \mid b] = \left(\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array}\right).$$

The vertical line separates the right sides from the coefficients. Its last column is column $n + 1$ and does not represent another unknown.

Linear algebra packages the unknown list into a column. We write such columns in italic type, while bold letters remain available for geometric vectors. The unknown column is

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in K^n,$$

where K^n is the Cartesian product of n copies of K. For this coefficient matrix, define Ax to be the column with entries $(Ax)_i = \sum_{j=1}^{n} a_{ij} x_j$ for $i = 1, \ldots, m$. Then $Ax = b$ is notation for (17.1). This matrix–column action uses the given coefficients. General matrix products are developed in Lecture 19.

17.2 Elementary row operations preserve solutions

Two systems in the same unknowns are *equivalent* if their solution sets are equal. Each row of the augmented matrix represents one whole equation, including its right side. Denote its current row by R_i. The *elementary row operations* are:

1. exchange two rows, $R_i \leftrightarrow R_j$;
2. multiply one row by a nonzero scalar $c \in K$;
3. replace R_i by $R_i + cR_j$, where $i \neq j$ and $c \in K$.

Every entry in the chosen row is changed, including its right side. The last operation leaves the other row R_j unchanged.

> **Theorem 17.1** (Solution preservation)**.** *Each elementary row operation produces an equivalent system. Every finite sequence of these operations therefore preserves the entire solution set.*

Proof. Exchanging equations changes only their order, not the requirement that all hold. Multiplying an equality by c preserves it, and division by the nonzero c recovers the original equality. Thus the first two operations work in both directions.

For row addition, every solution of the old system satisfies the sum of its row-i equation and c times its row-j equation. It therefore satisfies the new row-i equation and all unchanged equations. Conversely, the new system still includes the unchanged row-j equation. Subtracting c times it from the new row-i equation recovers the old one. This proves equality of solution sets. Applying this equality successively proves the finite-sequence statement. ■

Scaling a row by zero is excluded because it cannot be reversed. For instance, it would replace $x_1 = 1$ by $0 = 0$ and lose a restriction. A zero candidate cannot be used as a divisor. A row exchange may supply a nonzero entry instead.

17.3 Echelon form and the finite elimination procedure

A *zero row* has all entries zero, including the augmented entry when there is one. The *leading entry* of a nonzero row is its first nonzero entry from the left. A matrix is in *row-echelon form* if its zero rows are at the bottom and the leading entry of each successive nonzero row is strictly to the right of the leading entry in the row above. A *pivot* in an echelon matrix is such a leading entry. The echelon conditions imply zero entries below every pivot. A matrix is in *reduced row-echelon form* if it is in echelon form, every pivot is 1 and every other entry in a pivot column is zero.

> **Proposition 17.2** (Elimination)**.** *Every finite matrix over K can be put into row-echelon form and then reduced row-echelon form by elementary row operations.*

Proof. Start with the first row and column. Among the remaining rows, find the leftmost column with any nonzero entry. If no such column exists, all remaining rows are zero and the process stops. Otherwise exchange rows to put one of its nonzero entries in the next available row. This entry is the pivot. For each row below, subtract the pivot row multiplied by the quotient of that row's pivot-column entry by the nonzero pivot. The pivot-column entries below it become zero.

Repeat on the rows below and columns to the right. The unprocessed rows have zeros in the earlier pivot columns and in the columns that were skipped as zero. Subtracting the pivot row preserves these zeros because its entries to the left of its pivot are zero. Each step uses a new row and a strictly later column. There are finitely many rows and columns, so the process stops and has the required echelon conditions. This is *Gaussian elimination*.

To reduce further, start with the bottom pivot row and divide it by its nonzero pivot. Subtract multiples of it from all rows above to clear the pivot column. Continue upwards. A pivot row has zeros to the left of its pivot, so earlier leading positions are unchanged. Its entries in the later pivot columns have already been cleared, so the operation does not undo those cleared columns. After finitely many rows, every pivot is 1 and is the sole nonzero entry in its column. This is *Gauss–Jordan elimination*. ■

The proposition proves that a reduction can be performed. It does not yet claim that two routes have the same reduced form. For solving a given system, solution preservation holds along the route actually used.

17.4 One family with three possible outcomes

First see the row operations on a numerical system. For $x + y = 3$ and $2x + 3y = 8$, subtract twice the first row from the second:

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 3 & 8 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 2 \end{array}\right).$$

The second equation gives $y = 2$, and the first then gives $x = 1$. The same elimination below leaves a parameter in the final row, so we must examine whether that coefficient is zero before dividing by it.

Let $\lambda, \mu \in K$ and consider

$$\begin{aligned} x_2 + 3x_3 &= 6, \\ x_1 + 2x_2 - x_3 &= 3, \\ -x_1 - x_2 + (\lambda + 4)x_3 &= \mu + 3. \end{aligned} \tag{17.2}$$

Its augmented matrix is

$$\left(\begin{array}{ccc|c} 0 & 1 & 3 & 6 \\ 1 & 2 & -1 & 3 \\ -1 & -1 & \lambda+4 & \mu+3 \end{array}\right).$$

The first coefficient in the first row is zero. First exchange R_1 and R_2. Then replace R_3 by $R_3 + R_1$, using the new first row. These operations give

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & 3 & 6 \\ 0 & 1 & \lambda+3 & \mu+6 \end{array}\right).$$

Next replace R_3 by $R_3 - R_2$. The result is

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & 3 & 6 \\ 0 & 0 & \lambda & \mu \end{array}\right). \tag{17.3}$$

The last equation is $\lambda x_3 = \mu$. We have reached an echelon form in every parameter case. Its last row can lead in the third column, lead in the augmented column, or be a zero row. No division by λ has been made.

A nonzero last coefficient

If $\lambda \neq 0$, the last row determines $x_3 = \mu/\lambda$. The second row then determines x_2, and the first determines x_1:

$$x_3 = \frac{\mu}{\lambda}, \qquad x_2 = 6 - 3\frac{\mu}{\lambda}, \qquad x_1 = -9 + 7\frac{\mu}{\lambda}.$$

Every division here has a nonzero denominator. Every solution must have these values, and substitution in the three reduced equations verifies them. Reversibility then verifies the original system. The solution is unique.

For a numerical reduction, take $\lambda = \mu = -2$. In (17.3), divide R_3 by -2 to give pivot 1. Replace R_2 by $R_2 - 3R_3$. Finally replace R_1 by $R_1 + R_3$ and then by $R_1 - 2R_2$. The reduced form is

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 1 \end{array}\right).$$

It gives $x_1 = -2$, $x_2 = 3$ and $x_3 = 1$. For these parameters, the original third equation has coefficient $\lambda + 4 = 2$ and right side $\mu + 3 = 1$. The original equations check as

$$3 + 3 \cdot 1 = 6, \qquad -2 + 2 \cdot 3 - 1 = 3, \qquad -(-2) - 3 + 2 \cdot 1 = 1.$$

A zero last row

If $\lambda = \mu = 0$, the last equation is $0 = 0$ and imposes no restriction. Clear the second pivot above it by replacing R_1 by $R_1 - 2R_2$. The reduced form is

$$\left(\begin{array}{ccc|c} 1 & 0 & -7 & -9 \\ 0 & 1 & 3 & 6 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

The third column has no pivot. Its unknown is a *free variable*: it may be assigned arbitrarily, and the pivot variables are determined by that assignment. Put $x_3 = t$ with $t \in K$. Every solution column is then

$$x(t) = \begin{pmatrix} -9+7t \\ 6-3t \\ t \end{pmatrix}, \qquad t \in K. \tag{17.4}$$

The three original left sides become

$$(6-3t)+3t = 6, \qquad (-9+7t)+2(6-3t)-t = 3,$$

$$-(-9+7t)-(6-3t)+4t = 3.$$

They equal the required right sides for every t. Conversely, the reduced equations force the first two components once $x_3 = t$ is specified, so this family includes every solution. Different values of t give different third components. Since both $\mathbb{R}$ and $\mathbb{C}$ are infinite, there are infinitely many solutions.

A contradiction

If $\lambda = 0$ and $\mu \neq 0$, the last row is the equation $0 = \mu$. It cannot be satisfied by any unknowns. Solution preservation shows that the original system is inconsistent. A nonzero augmented entry after zero coefficient entries is a contradiction, whereas an entirely zero row is an identity. These two row types have different effects even though both have zero coefficients of the unknowns.

17.5 The consistency criterion and all free parameters

In any echelon form of an augmented matrix, call a column among the first n a *pivot column* if it contains a pivot. Once contradictions have been excluded, each nonpivot unknown is called free. Give different free variables separate parameters. Let r be the number of coefficient pivot columns in this particular reduction. There are $n-r$ nonpivot columns.

> **Theorem 17.3** (Consistency and free-variable criterion)**.** *For an echelon form obtained from $[A \mid b]$ by row operations, the system is consistent exactly when the augmented column has no pivot. Equivalently, the coefficient block and the full augmented matrix in that form have equally many pivots. If consistent, its solutions are parameterised by arbitrary values of the $n-r$ free variables. It has one solution when $r = n$ and infinitely many when $r < n$.*

Proof. A pivot in column $n+1$ is the first nonzero entry of its row. All preceding coefficient entries are zero, so that row is $0 = c$ with $c \neq 0$. It prevents any solution. Conversely, if the augmented column has no pivot, every row with zero coefficients also has zero right side. Reduce the echelon matrix further by Gauss–Jordan operations. These operations preserve the pivot positions as shown in the elimination proof, so there is still no augmented pivot.

Choose arbitrary values for each free unknown. In each nonzero reduced row, its pivot is 1 and the entries in all other pivot columns are zero. That row determines its one pivot unknown by subtracting the free-variable terms from its right side. Every row is now satisfied, and zero rows read $0 = 0$. This constructs a solution for every free assignment. Every solution must satisfy the same pivot equations, so no further solutions exist and the assignment determines the pivot entries uniquely.

If there are no free variables, the construction gives exactly one solution. If there is a free variable, varying it through K changes that component and hence gives infinitely many distinct solutions. Finally, removing the augmented column removes one pivot precisely when that column contains a pivot. This proves the equivalent pivot-count criterion in the actual form under consideration. ■

The theorem includes a matrix with no coefficient pivots. If its right sides also vanish, every unknown is free and every column $x \in K^n$ is a solution. A nonzero right side instead produces a contradiction. The number of equations alone does not settle consistency. For the family (17.2), the complete classification is

parameters	last equation	solution set
$\lambda \neq 0$	$\lambda x_3 = \mu$	one solution
$\lambda = 0,\ \mu = 0$	$0 = 0$	family (17.4)
$\lambda = 0,\ \mu \neq 0$	$0 = \mu$	$\varnothing$

17.6 Uniqueness of reduced form

Theorem 17.4 (Uniqueness of reduced form · *cited*)**.** *A matrix over $\mathbb{R}$ or $\mathbb{C}$ has a unique reduced row-echelon form under row operations. In particular, its pivot columns and pivot count are independent of the elimination route.*

The uniqueness result is used without proof; see Lay, Lay and McDonald, §1.2, Theorem 1. The connection to pivot counts is that normalising pivots and clearing above them does not move their columns. Every echelon form therefore has the same pivot columns as its reduced form. Uniqueness makes those columns, and hence their number, independent of the starting choices during elimination. This route-independence statement retains the cited status.

The earlier consistency proof applies to any actual reduction and did not use uniqueness of reduced form. Lecture 18 will use the route-independent coefficient pivot count to define rank and a dimension convention for the homogeneous solution set. Its dependence on this cited result will be explicit.

Summary 17.

- Equation (17.1): a linear system is encoded by its coefficient matrix and right-side column. The augmented column represents constants, not a further unknown.
- Theorem 17.1: the three reversible row operations preserve the whole solution set. Scaling and pivot division require nonzero scalars.
- Proposition 17.2: Gaussian and Gauss–Jordan elimination terminate in echelon and reduced echelon forms.
- Theorem 17.3: an augmented pivot is exactly a contradiction. Otherwise each free assignment determines exactly one solution, giving one solution with no free variable and infinitely many when one is present.
- Theorem 17.4 · *cited*: uniqueness of reduced form and route-independent pivot counts are separate cited facts.

Sheet 17 tests elimination, nontrailing free columns and parameter cases. Lecture 18 describes the complete solution set using a particular solution and solutions of the system with zero right sides.

References

D. C. Lay, S. R. Lay and J. J. McDonald, *Linear Algebra and Its Applications*, 5th ed., Pearson, 2016, §§1.1–1.2. S. Lang, *Introduction to Linear Algebra*, 2nd ed., Springer, 1986, Chapter II, §4.